

Hany Jiang

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Education

University of Waterloo, Data Science, Bachelor of Mathematics

Sep 2024 – Dec 2028

Work Experience

Technology Operations Developer – Instant Risk Coverage

May 2026 – Aug 2026

- Enhanced more than 10 Mage AI ETL pipelines transforming Medusa e-commerce data through **PostgreSQL** staging into data mart tables powering Metabase reporting.
- Built the end-to-end reporting pipeline for a new insurance platform, landing an internal pricing service into a new staging layer and **data mart**, then calculating premium, commission, and net figures at separate insurer and broker rates to generate 256 monthly **Excel** statements sent to carriers.
- Automated monthly policy-level premium reporting, joining order, line-item, customer, entity, SKU, and payment data in **SQL** with **Python** validation and formatting, and reconciling output against the legacy system before cutover.
- Built a **Node.js** and **MongoDB** toolkit processing parent-child policy cancellations with rollback snapshots, edge-case validation, and **CSV-to-JSON** conversion, cutting processing time by over **90%**.

Data Analyst – Christ City Ministries

Sep 2025 – Dec 2025

- Cleaned and standardized 10+ Ontario demographic datasets from secondary research and public sources, producing analysis-ready tables to benchmark regional population trends across 15 municipalities.
- Built an interactive **Power BI** dashboard with a relational data model and custom DAX measures, enabling stakeholders to explore community needs, attendance trends, and participation rates with drill-down filtering.

Projects & Research

FoKy – iOS Productivity & Study App

In Development

- Developed a production-oriented iOS application using **React Native**, **Expo**, and **TypeScript**, combining app blocking, timed focus sessions, flashcard review, and collectible rewards.
- Reduced hallucination risk by building a privacy-conscious study agent with **Claude** that transforms notes and selected PDF pages into source-grounded flashcards using structured outputs, deterministic validation, semantic quality checks, prompt-injection defenses, and human approval.
- Architected a secure backend using **Supabase**, **PostgreSQL**, Row-Level Security, transactional RPCs, and 57 **Edge Functions** to manage authentication, progress, rewards, subscriptions, and social features.
- Improved reliability through idempotent APIs, retryable outboxes, offline synchronization, and concurrency safeguards, validated by 500+ **Jest** tests, 90 **Deno** tests, database security tests, and automated production smoke tests.

Multitask Learning for Cross-Asset Price Prediction – Directed Research Project

GitHub | Presentation

- Built an end-to-end financial time-series pipeline using **Python**, **Pandas**, **NumPy**, and **Matplotlib** for cross-asset alignment, missing-value imputation, normalization, sliding-window construction, and technical-indicator feature engineering.
- Designed and evaluated linear regression, feedforward neural network, and multitask-learning models using **PyTorch**; the multitask model reduced MSE by **42.9%** and runtime by **89.5%** compared with the single-task FNN, while revealing negative transfer and limited-data constraints.

NASA Space Apps Hackathon: SPIKE – AI Research Dashboard

GitHub | NASA Space Apps

- Architected a full-stack research platform using **React**, **Next.js**, and **Flask**, enabling rapid querying of 600+ NASA bioscience publications.
- Implemented an automated ETL pipeline using **Python** and the **OpenAI API**, applying strict **JSON Schema** validation to transform unstructured research text into type-safe, queryable datasets.

Technical Skills

- Languages:** C++, C, Python, TypeScript, JavaScript, SQL, Bash, R
- Frameworks:** React, React Native, Next.js, Node.js, Flask, Expo
- AI & Data:** PyTorch, OpenAI API, Anthropic Claude API, Pandas, NumPy, Matplotlib, JSON Schema
- Databases & Cloud:** PostgreSQL, MongoDB, SQLite, Supabase, Cloudflare
- Testing & DevOps:** Jest, Deno, Git, Linux, Docker, GitHub Actions, Jira